# Sardar Patel Institute of Technology

**Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India**
**(Autonomous Institute Affiliated to University of Mumbai)**

**ESE Examination**
December 2024

Max. Marks: 100 Duration: 3Hrs.
Class: T.E. Semester-V Computer Engineering Course Code: CS302
Name of the Course: Software Engineering

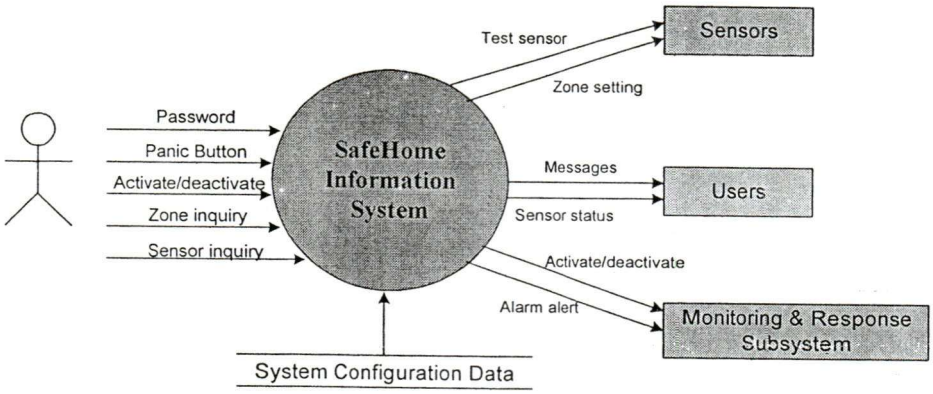
Instructions:
(1) All questions are compulsory
(2) Draw neat diagrams
(3) Assume suitable data if necessary

Course Outcomes: At the end of the course student will be able to:
(1) Analyze software requirements.
(2) Apply UML models for a project.
(3) Evaluate system architecture and develop detailed task schedule from the overall estimates and planning.
(4) Illustrate different coding principles with unit test process.
(5) Understand the need for DevOps.

| Q. No. | | Max Marks | CO-BL |
|---|---|---|---|
| Q.1 A) | Illustrate agile SDLC model XP with diagram. Also, briefly explain the terms pair programming, test driven development, and refactoring. | 10 | CO1-2 |
| Q.1 B) | Prepare Software requirement specification for Moodle a Learning Management System. | 10 | CO1-3 |
| Q.2 A) | Write all notations along with their semantics which are used in a Class Diagram. Draw Class diagram for online shopping domain model.<br>OR<br>Draw use case diagram for First Year Engineering Admission System | 10 | CO2-3 |
| Q.2 B) | A student wants to issue a book from library. At first the student requests to the librarian for a book which he/she needs. The librarian searches a book in the database. The book database system searches the book availability and checks the book availability and checks the details of the book. The database system returns the details of the book to the librarian. Then the librarian requests library card from the student. The student provides the library card, and the librarian updates the details into the database. After completion of entering the details in the database. A receipt is generated for the librarian. Finally, a book is issued to the students. Based on the details of the above scenario, Draw sequence and collaboration diagram. | 10 | CO2-3 |

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| **Q.3 A)** | Suppose you are the project manager of a software project requiring the following activities<br><br>Show Critical Path and Compute Slack time (ST) for each task. | 10 | CO3-3 |

| Task Name | Task | Duration | Precedence |
|---|---|---|---|
| A | Hardware Selection | 5 | -- |
| B | System Configuration | 6 | -- |
| C | Install Hardware | 4 | A |
| D | Data Migration | 3 | A |
| E | Draft Office Procedures | 1 | A |
| F | Recruit Staff | 4 | E |
| G | User Training | 14 | D,F |
| H | Install and Test | 12 | B,C |
| I | Maintenance | 2 | G,H |

| | | | |
|---|---|---|---|
| **Q.3 B)** | Prepare risk information sheet for (i) Project risk, (ii) Technical risk (iii) Business risk (Assume suitable risks) | 10 | CO3-3 |
| **Q.4 A)** | For Safe home Information System estimate the effort and cost required using Function Point (FP). Suppose the organizational average productivity for system of this type =1.2 FP/PM and the cost per function point is Rs. 20,000. Assume all the data are of simple complexity.<br><br>  <br><br><br>OR<br>Prepare structured analysis DFD (level 0,1, and level 2) and structured design (structure Chart) for developing the Restaurant Automation System | 10 | CO3-3 |
| **Q.4 B)** | Design equivalence class test cases for<br>(i) A program that reads two integer pairs (m1, c1) and (m2, c2) defining two straight lines of the form y=mx+c. The program computes the intersection point of the two straight lines and display the point of intersection.<br>(ii) A function that reads a character string a character string of size lesss than five characters and displays whether it is a palindrome. | 5+5 | CO4-3 |

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| **Q.5 A)** | (i) Consider the following function. Draw Control flow graph. From the control flow graph, determine its Cyclomatic complexity and find linearly independent paths. Design test cases for the function.<br><br>int f1(int x,int y)<br>{<br>while (x != y)<br>{<br>if (x>y) then<br>x=x-y;<br>else y=y-x;<br>}<br>return x;<br>}<br><br>(ii) Suppose in order to estimate the number of latent errors in a program, you seed it with 1000 errors of different kinds. After testing the software using its full test suite, you discover only 500 of the seeded errors. You discover 250 other errors also. Estimate the number of latent errors in the software | 5+5 | CO4-3 |
| **Q. 5 B)** | Discuss the working of DevOps with suitable example?<br>OR<br>Illustrate Software Configuration management process. | 10 | CO5-2 |

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